

Synthetic data as vicinal gradient coverage: why smoothness improves while margin falls

Mechanism theory for the diffusion-data robustness finding (author-derived; verified)

June 29, 2026

Abstract

The pilot observation is that adding diffusion-generated data to adversarial training increases the robust radius even though the raw logit margin *decreases*, because the local sensitivity $L = \mathbb{E}\|\nabla_x M\|$ falls by more than the margin. This note gives a static mechanism for that pattern. First-order adversarial training is input-gradient regularization (Lemma 1); synthetic data changes the measure on which the penalty is enforced, adding a positive-semidefinite *vicinal gradient-coverage matrix*; increasing that matrix suppresses high-gradient modes (Theorem 1); and if a high-gradient mode also carries raw margin, the margin falls while the sensitivity falls faster, so the scale-invariant radius proxy M/L rises exactly when a smoother low-gradient component remains (Theorem 2). A predicted signature, $L_1/L_2 \approx \text{const}$ while both fall (Corollary 2), is already confirmed in the pilot ($25.4 \rightarrow 23.6$). The slogan: *diffusion data suppresses high-gradient margin-carrying modes; the raw margin falls, but the sensitivity falls faster, so M/L rises.*

The robust radius admits the first-order decomposition $\Delta \log r \approx \Delta \log M - \Delta \log L$, so a radius gain with a smaller margin is possible exactly when L falls by more than M . The mechanism is not that synthetic data raises the margin: it is that adversarial training is, to first order, input-gradient regularization, and synthetic data changes the measure on which this regularizer acts, adding a positive-semidefinite vicinal gradient-coverage matrix. Modes that carry margin but have high input-gradient energy are suppressed; the raw margin can fall while the sensitivity falls faster, and the scale-invariant proxy M/L rises.

1 First-order adversarial training is input-gradient regularization

Lemma 1 (First-order robust-loss expansion). *Let $\ell : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable on the closed ball $B_\varepsilon(x) = \{x + \delta : \|\delta\| \leq \varepsilon\}$ for a norm $\|\cdot\|$ with dual $\|\cdot\|_*$, and suppose $\|\nabla_x^2 \ell(z)\|_{\text{op}} \leq H$ for all $z \in B_\varepsilon(x)$ (operator norm induced by $\|\cdot\|$). Then*

$$\sup_{\|\delta\| \leq \varepsilon} \ell(x + \delta) = \ell(x) + \varepsilon \|\nabla \ell(x)\|_* + R_\varepsilon(x), \quad |R_\varepsilon(x)| \leq \frac{H}{2} \varepsilon^2.$$

Proof. Set $g = \nabla \ell(x)$. For $\|\delta\| \leq \varepsilon$, Taylor with remainder gives $\ell(x + \delta) = \ell(x) + \langle g, \delta \rangle + \frac{1}{2} \delta^\top \nabla^2 \ell(x + s\delta) \delta$ for some $s \in [0, 1]$, and $|\frac{1}{2} \delta^\top \nabla^2 \ell(x + s\delta) \delta| \leq \frac{H}{2} \|\delta\|^2 \leq \frac{H}{2} \varepsilon^2$. With $\langle g, \delta \rangle \leq \|g\|_* \|\delta\| \leq \varepsilon \|g\|_*$, the supremum obeys $\sup_{\|\delta\| \leq \varepsilon} \ell(x + \delta) \leq \ell(x) + \varepsilon \|g\|_* + \frac{H}{2} \varepsilon^2$. For the lower bound pick v with $\|v\| \leq 1$ and $\langle g, v \rangle = \|g\|_*$ (the unit ball is compact, so the linear functional attains its max), and set $\delta = \varepsilon v$. Taylor gives $\ell(x + \varepsilon v) \geq \ell(x) + \varepsilon \|g\|_* - \frac{H}{2} \varepsilon^2$, and the supremum is at least this. Combining the two bounds gives $|\sup_\delta \ell(x + \delta) - \ell(x) - \varepsilon \|g\|_*| \leq \frac{H}{2} \varepsilon^2$. $\square$

Remark 1 (From loss-gradient to margin-gradient). If $\ell(x, y) = \phi(M(x, y))$ then $\nabla_x \ell = \phi'(M) \nabla_x M$, and on any margin band where $0 < c_- \leq |\phi'(M)| \leq c_+ < \infty$ the loss-gradient and margin-gradient penalties are equivalent up to constants: $c_- \|\nabla_x M\| \leq \|\nabla_x \ell\| \leq c_+ \|\nabla_x M\|$. Thus the first-order adversarial penalty controls the same local sensitivity $L = \mathbb{E} \|\nabla_x M\|$ that enters the radius proxy.

2 The vicinal gradient-coverage matrix

Linearize the margin in parameters, $M_\theta(x) = \theta^\top \psi(x)$ with a differentiable feature map $\psi : \mathcal{X} \rightarrow \mathbb{R}^p$, and write $J_\psi(x) = \nabla_x \psi(x) \in \mathbb{R}^{p \times d}$, so that $\nabla_x M_\theta(x) = J_\psi(x)^\top \theta$ and $\|\nabla_x M_\theta(x)\|_2^2 = \theta^\top J_\psi(x) J_\psi(x)^\top \theta$. For a (data or vicinal) measure ν define the gradient-coverage matrix

$$Q_\nu := \int J_\psi(x) J_\psi(x)^\top d\nu(x), \quad \int \|\nabla_x M_\theta(x)\|_2^2 d\nu(x) = \theta^\top Q_\nu \theta.$$

Q_ν is positive semidefinite, since $v^\top Q_\nu v = \int \|J_\psi(x)^\top v\|_2^2 d\nu(x) \geq 0$. Adding synthetic data changes ν , adding a positive-semidefinite Q_{syn} to the input-gradient penalty.

3 Spectral rough-mode filtering

Theorem 1 (Synthetic gradient coverage filters high-sensitivity modes). *Let $A \in \mathbb{R}^{p \times p}$ be symmetric positive definite, $Q \in \mathbb{R}^{p \times p}$ symmetric positive semidefinite, $a \in \mathbb{R}^p$, and for $t \geq 0$ set $B_t = A + tQ$ and $\mathcal{J}_t(\theta) = \frac{1}{2} \theta^\top B_t \theta - a^\top \theta$ with minimizer θ_t . With $\eta(t) = a^\top \theta_t$ and $S(t) = \theta_t^\top Q \theta_t$,*

$$\eta'(t) = -S(t) \leq 0, \quad S'(t) = -2\theta_t^\top Q B_t^{-1} Q \theta_t \leq 0.$$

Proof. B_t is symmetric positive definite for $t \geq 0$, so $\mathcal{J}_t$ is strictly convex with unique minimizer $\theta_t = B_t^{-1} a$ ($B_t \theta_t = a$). Differentiating $B_t \theta_t = a$ in t and using $B'_t = Q$ gives $Q \theta_t + B_t \theta'_t = 0$, i.e. $\theta'_t = -B_t^{-1} Q \theta_t$. Then $\eta'(t) = a^\top \theta'_t = (B_t \theta_t)^\top \theta'_t = \theta_t^\top B_t \theta'_t = \theta_t^\top (-Q \theta_t) = -S(t)$, and $S(t) = \theta_t^\top Q \theta_t \geq 0$. Also $S'(t) = 2(\theta'_t)^\top Q \theta_t = 2(-B_t^{-1} Q \theta_t)^\top Q \theta_t = -2\theta_t^\top Q B_t^{-1} Q \theta_t = -2z^\top B_t^{-1} z$ with $z = Q \theta_t$, and $z^\top B_t^{-1} z \geq 0$ since $B_t^{-1} \succ 0$. $\square$

4 A two-mode theorem: margin down, sensitivity down more, radius up

Theorem 2 (Smoothness-not-margin in a two-mode model). *Let $\theta = (\theta_s, \theta_r) \in \mathbb{R}^2$ (smooth mode s , rough mode r), $A = I_2$, $Q = \begin{pmatrix} 0 & 0 \\ 0 & q \end{pmatrix}$ with $q > 0$, and $a = (a_s, a_r)$ with $a_r \neq 0$. For $t \geq 0$ let θ_t minimize $\frac{1}{2} \theta^\top (I_2 + tQ) \theta - a^\top \theta$, and set $\eta(t) = a^\top \theta_t$, $L(t) = \sqrt{\theta_t^\top Q \theta_t}$, $\rho(t) = \eta(t)/L(t)$. Then*

$$\theta_t = \left(a_s, \frac{a_r}{1+tq} \right), \quad \eta(t) = a_s^2 + \frac{a_r^2}{1+tq}, \quad L(t) = \frac{\sqrt{q} |a_r|}{1+tq}, \quad \rho(t) = \frac{(1+tq)a_s^2 + a_r^2}{\sqrt{q} |a_r|},$$

$$\eta'(t) = -\frac{qa_r^2}{(1+tq)^2} < 0, \quad L'(t) = -\frac{q\sqrt{q} |a_r|}{(1+tq)^2} < 0, \quad \rho'(t) = \frac{\sqrt{q} a_s^2}{|a_r|} > 0 \quad (a_s \neq 0),$$

and, whenever $a_s \neq 0$, $-\frac{d}{dt} \log L(t) > -\frac{d}{dt} \log \eta(t)$. The raw margin decreases, the sensitivity decreases faster, and the radius proxy η/L strictly increases.

Proof. $(I_2 + tQ) = \text{diag}(1, 1 + tq)$, so $(I_2 + tQ)\theta_t = a$ gives $\theta_{s,t} = a_s$ and $\theta_{r,t} = a_r/(1 + tq)$. Then $\eta(t) = a_s \cdot a_s + a_r \cdot \frac{a_r}{1+tq} = a_s^2 + \frac{a_r^2}{1+tq}$. $Q\theta_t = (0, qa_r/(1 + tq))^\top$, so $\theta_t^\top Q\theta_t = \frac{a_r}{1+tq} \cdot \frac{qa_r}{1+tq} = \frac{qa_r^2}{(1+tq)^2}$ and $L(t) = \sqrt{q}|a_r|/(1 + tq)$ (as $1 + tq > 0$). Dividing, $\rho(t) = (a_s^2 + \frac{a_r^2}{1+tq}) \cdot \frac{1+tq}{\sqrt{q}|a_r|} = \frac{(1+tq)a_s^2 + a_r^2}{\sqrt{q}|a_r|}$. Differentiating: $\eta'(t) = a_r^2 \cdot (-q)(1 + tq)^{-2} = -\frac{qa_r^2}{(1+tq)^2} < 0$; $L'(t) = \sqrt{q}|a_r| \cdot (-q)(1 + tq)^{-2} = -\frac{q\sqrt{q}|a_r|}{(1+tq)^2} < 0$; the numerator of ρ has t -derivative qa_s^2 over the constant denominator, so $\rho'(t) = \frac{qa_s^2}{\sqrt{q}|a_r|} = \frac{\sqrt{q}a_s^2}{|a_r|} > 0$ when $a_s \neq 0$. For the logarithmic comparison, $\log L(t) = \log(\sqrt{q}|a_r|) - \log(1 + tq)$ gives $-\frac{d}{dt} \log L = \frac{q}{1+tq}$, and $-\frac{d}{dt} \log \eta = \frac{qa_r^2}{(1+tq)^2 \eta(t)}$. The inequality $\frac{q}{1+tq} > \frac{qa_r^2}{(1+tq)^2 \eta(t)}$ is, after multiplying by $(1 + tq)^2 \eta(t)/q > 0$, equivalent to $(1 + tq)\eta(t) > a_r^2$, i.e. $(1 + tq)a_s^2 + a_r^2 > a_r^2$, i.e. $(1 + tq)a_s^2 > 0$, i.e. $a_s \neq 0$. $\square$

Corollary 1 (Pure temperature shrinkage is the degenerate case). *If $a_s = 0$ in Theorem 2, then $\rho(t) = |a_r|/\sqrt{q}$ is constant: suppressing only a single rough margin-carrying mode scales margin and sensitivity together and cannot improve the radius proxy. A genuine radius gain requires a remaining smooth component $a_s \neq 0$.*

Proof. Set $a_s = 0$: $\rho(t) = \frac{a_r^2}{\sqrt{q}|a_r|} = \frac{|a_r|}{\sqrt{q}}$, independent of t , so $\rho'(t) = 0$. $\square$

Corollary 2 (Uniform gradient-shape shrinkage). *If the gradient field is dominated by the rough mode, $\nabla_x M_{\theta_t}(x) \approx \theta_{r,t} g_r(x)$ with g_r independent of t , then for any two homogeneous norms $\|\cdot\|_\alpha, \|\cdot\|_\beta$,*

$$\frac{\|\nabla_x M_{\theta_t}(x)\|_\alpha}{\|\nabla_x M_{\theta_t}(x)\|_\beta} \approx \frac{\|g_r(x)\|_\alpha}{\|g_r(x)\|_\beta},$$

approximately independent of t : L_1 and L_2 can both drop sharply while L_1/L_2 stays nearly constant.

Proof. Homogeneity gives $\|\nabla_x M_{\theta_t}(x)\|_\alpha \approx |\theta_{r,t}| \|g_r(x)\|_\alpha$ and likewise for β ; since $\theta_{r,t} = a_r/(1 + tq) \neq 0$, $|\theta_{r,t}|$ cancels in the ratio. $\square$

5 Interpretation

The smooth mode s is a low-sensitivity component that generalizes across the vicinity of the data manifold; the rough mode r is a high-sensitivity component that can raise the raw logit margin on the finite real training set. Synthetic data increases the vicinal coverage tq on the rough mode, so $\theta_{r,t} = a_r/(1 + tq)$ shrinks: the raw margin $\eta(t) = a_s^2 + a_r^2/(1 + tq)$ falls (the rough mode contributed margin), but the sensitivity $L(t) = \sqrt{q}|a_r|/(1 + tq)$ falls faster (it is entirely the rough mode), so the radius proxy $\rho(t)$ rises whenever $a_s \neq 0$. A pure logit-temperature rescaling would shrink margin and sensitivity by the same factor and leave M/L unchanged; that is the degenerate $a_s = 0$ case. The observed increase in M/L corresponds to $a_s \neq 0$: a smooth component remains while synthetic gradient coverage suppresses rough sensitivity.

6 Adversarial verification

Remark 2 (Scope and falsifiable predictions). **(1) Not every synthetic data helps.** A useful radius gain needs Q to penalize a rough mode more than a smooth margin-carrying one; if $Q = qI$ all modes scale uniformly and the mechanism degenerates to temperature-like shrinkage (no forced gain). **(2) It explains margin shrink, not denies it:** $\eta'(t) = -S(t) \leq 0$. **(3) It**

explains L falling faster than M : $-\frac{d}{dt} \log L > -\frac{d}{dt} \log \eta$ iff $a_s \neq 0$, exactly the condition for η/L to rise while η falls. **(4) It is not a robust-overfitting theorem:** the proof uses no time/epochs/learning-rate decay; it is a static regularization-path result. If the effect appears early and varies monotonically with synthetic diversity, this is the mechanism; if it appears only after a late robust-overfitting phase, a separate dynamical theorem is needed. **(5) Local, not global:** the quadratic model is a local linearization; the theorem identifies the algebraic condition for the observed decomposition, not a claim that every diffusion-trained net optimizes this exact objective. **(6) Falsifiable predictions:** L decreases monotonically with synthetic diversity; M may decrease; M/L increases when a smooth component remains; $L_1/L_2 \approx \text{const}$ when the same rough gradient shape is suppressed in amplitude (Corollary 2); and the largest L drop should occur on vicinal/interpolation points where the new synthetic coverage differs most from the empirical measure.

Empirical status (pilot). The monotone predictions hold: L_2 falls monotonically with synthetic amount ($5.20 \rightarrow 2.54$), M falls ($4.42 \rightarrow 2.58$), M/L rises, and the gradient-shape prediction $L_1/L_2 \approx \text{const}$ is confirmed ($25.4 \rightarrow 23.6$ while L_1, L_2 each roughly halve). The decisive remaining test is the vicinal-concentration prediction (largest L drop on/near synthetic points), the **manifold** measurement.